Abhinav Gorantla

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Education

Arizona State University, *School of Computing and Augmented Intelligence* *Jan 2026 – Present*
Ph.D. in Computer Science

- **Advisor:** Dr. K. Selçuk Candan.
- **Research areas:** Causally informed Machine Learning, Multi-Objective Optimization.

Arizona State University, *School of Computing and Augmented Intelligence* *Aug 2023 – Dec 2025*
Master of Science in Computer Science, GPA: 4.0/4.0

- **Honors:** With Distinction.
- **Thesis:** *Speeding-up of the Non-Dominated Sorting Genetic Algorithm-II by Selective De-Correlation*
- **Advisor:** Dr. K. Selçuk Candan.

Vellore Institute of Technology, *School of Computer Science and Engineering* *Jul 2019 – May 2023*
Bachelor of Technology in Computer Science and Engineering

- **GPA:** 8.94/10.0

Publications

Causal Search for Skylines (CSS): Causally-Informed Selective Data De-Correlation
Pratanu Mandal, ***Abhinav Gorantla***, K. Selçuk Candan, Maria Luisa Sapino
Accepted to ACM SIGMOD 2026.
Proposes a method for leveraging causal structure to selectively de-correlate data, improving efficiency of Skyline retrieval in relational databases.

Preprint: arxiv.org/abs/2603.14339
Code: github.com/EmitLab/Causal-Search-for-Skylines

CausalBench-ER: Causally-Informed Explanations and Recommendations for Reproducible Benchmarking
Ahmet Kapkiç, Pratanu Mandal, ***Abhinav Gorantla***, Shu Wan, Ertugrul Çoban, Paras Sheth, Huan Liu, K. Selçuk Candan
In Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25), 2025.
Extends CausalBench with a recommendation engine that suggests appropriate causal analysis pipelines based on dataset, model and hyperparameter characteristics.

DOI: 10.1145/3746252.3761606

Introducing CausalBench: A Flexible Benchmark Framework for Causal Analysis and Machine Learning *[Best Demo Award]*
Ahmet Kapkiç, Pratanu Mandal, Shu Wan, Paras Sheth, ***Abhinav Gorantla***, Yoonhyuk Choi, Huan Liu, K. Selçuk Candan
In Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM '24), 2024.
An open benchmarking platform for causal learning and machine learning.

DOI: 10.1145/3627673.3679218

Tutorials

CausalBench: Causal Learning Research Streamlined
Ahmet Kapkiç, Pratanu Mandal, ***Abhinav Gorantla***, Shu Wan, Ertugrul Çoban, Paras Sheth, Huan Liu, K. Selçuk Candan
Tutorial at the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2025), 2025.
DOI: 10.1145/3711896.3737598

Research and Professional Experience

Graduate Research Assistant — *Tempe, AZ*
EMIT Lab, Arizona State University — *August 2024 - (present)*

- **Advisor:** Dr. K. Selçuk Candan.
- **Research areas:** Causally informed Machine Learning, Multi-Objective Optimization.
- Working on causally informed training methods for Prior Fitted Networks (PFNs).
- Developing an optimized algorithm for efficient Skyline retrieval in relational database systems by leveraging causal structure in the data.
- Conducting research on optimizing the selection process in multi-objective evolutionary algorithms, specifically Pareto-based methods like NSGA and SPEA.
- Collaborating with researchers at CASCADE Lab to maintain and improve `causalbench.org`, a platform dedicated to benchmark causal learning algorithms.
- Collaborated with researchers at CASCADE Lab in developing a novel causally informed recommendation system for `causalbench.org`.

Graduate Services Assistant — *Tempe, AZ*
EMIT Lab, Arizona State University — *March 2024 - August 2024*

- **Advisor:** Dr. K. Selçuk Candan.
- **Research areas:** Causally informed Machine Learning.
- Supported CASCADE Lab researchers in developing the causalbench-asu Python package and website, establishing an end-to-end benchmarking solution for the causal machine learning community.
- Served as a full stack developer on the CausalBench project, contributing to a comprehensive framework for benchmarking causal machine learning algorithms.

Software Development Engineer Intern — *Bengaluru, India*
Webknot Technologies Pvt. Ltd. — *April 2022 - June 2023*

- Revamped API endpoints within the Palette project, achieving a notable 30% reduction in response times.
- Engineered a custom plugin for Sisense BI software, enabling the seamless display of geojson data on a GeoJSON layer atop maps rendered via DeckGL.
- Fine-tuned data flow for the DeckGL plugin within Sisense by elevating the efficiency of JAQL queries, ensuring a smoother and more responsive user experience.

Projects

Enhancing Diversity in the LLM Modulo Framework through Multi-Response Generation

- Developed the Diversified LLM Modulo framework to address looping and redundancy in the LLM Modulo framework.
- Improved the performance of the LLM Modulo Framework on Planning tasks. Tested my framework on the Google Deepmind Natural Plan benchmark (paper) and achieved a performance improvement of 300% by increasing the diversity of LLM (Large Language Model) Responses.
- Tools Used: Python, OpenAI API.

Research Publications Analysis tool

- Architected and developed a research publication analysis web application for ASU, integrating AWS S3, SageMaker, and SES. This tool leverages SCOPUS APIs to fetch research papers and perform text analysis of abstracts.

- Utilized LLM (GPT-4) for advanced feature extraction from scraped text.
- Optimized back-end system performance by reducing server response time by 80% and production costs by 40% through the strategic integration of AWS SageMaker.
- Tools Used: ReactJS, NodeJS, Python-FastAPI, RabbitMQ, MongoDB, AWS S3, AWS Sagemaker, OpenAI API.

Teaching Experience

ASU SCience and ENgineering Experience (SCENE) — Arizona State University
Co-mentored a SCENE high school student on event detection in multivariate time-series data; supported experimental design and preparation for AZSEF. — Spring 2026

Graduate Teaching Assistant — Arizona State University
CSE 510 Database Management Systems Implementation — Spring 2025

Graduate Teaching Assistant — Arizona State University
CSE 515 Multimedia and Web Databases — Fall 2024

Community Service

Student Volunteer
ACM Knowledge Discovery and Data Mining (KDD) 2025 Conference

Awards

University Graduate Fellowship 2026, School of Computing and AI
Arizona State University

Best Demo Award, CIKM 2024
Introducing CausalBench: A Flexible Benchmark Framework for Causal Analysis and Machine Learning

MS in CS with Distinction
Arizona State University, December 2025

Skills

Machine Learning: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn.

Programming: Python, C++, C, Java, TypeScript.

Databases: MongoDB, SQL, SQLAlchemy.

Cloud & Infrastructure: AWS (S3, EC2, SageMaker, Lambda), Google Firestore, Firebase Functions.

Web & Backend: Node.js, ReactJS, FastAPI, RabbitMQ.

Other: Git.

Leadership

Vice Chairperson, THEP Journalism Club, VIT (2021–2022)
Led technical development of the club's newsletter platform (MERN stack) and launched a podcast reaching ∼800 listeners in its first month.

Editor-in-Chief, Fifth Pillar NGO, VIT (2021–2022)
Directed a 50-member editorial team and introduced new content formats including "Law Talks," expanding social media and community outreach by 50%.

[0] References available upon request.